

Adaptive Sampling Strategy

Round 1: All Problems

Generate $n=3$ CoT traces per problem at $T=0.3$, compute avg log-probability

**HIGH
CONFIDENCE**
Keep all $n=3$

Confidence Check

$\max(\text{avg log-probability}) \geq -1.8 ?$

NO

Round 2+: Low-Confidence Problems Only

Iterative high-temp resampling: $T=0.9$, $n=2/\text{iter}$, up to 4 rounds

Re-check confidence each round; exit when threshold met or budget exhausted

Advantage: log-probability is returned natively by vLLM (logprobs=1) — zero additional inference cost